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Out With the Old and in With the New? An Empirical Comparison of Supervised Learning Algorithms to Predict Recidivism

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Abstract

Recent research has produced mixed results as to whether newer machine learning algorithms outperform older, more traditional methods such as logistic regression in predicting recidivism. In this study, we compared the performance of 12 supervised learning algorithms to predict recidivism among offenders released from Minnesota prisons. Using multiple predictive validity metrics, we assessed the performance of these algorithms across varying sample sizes, recidivism base rates, and number of predictors in the data set. The newer machine learning algorithms generally yielded better predictive validity results. LogitBoost had the best overall performance, followed by Random forests, MultiBoosting, bagged trees, and logistic model trees. Still, the gap between the best and worst algorithms was relatively modest, and none of the methods performed the best in each of the 10 scenarios we examined. The results suggest that multiple methods, including machine learning algorithms, should be considered in the development of recidivism risk assessment instruments.

Keywords

machine learning, predictive discrimination, calibration, recidivism, risk assessment

Introduction

The principles of effective correctional intervention hold that resources are used most effectively when they address the risk, needs, and responsivity of offenders (Cullen &

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Gendreau, 2000). According to the risk principle, correctional interventions should be reserved for the highest risk offenders so as to maximize the recidivism reduction from effective programming. By helping identify which areas to target for programming, the needs principle assumes that interventions that target an offender's criminogenic needs (dynamic risk factors) are more likely to decrease recidivism because changes can be made in these factors. And the responsivity principle suggests that interventions are more successful when they are tailored to their individual learning styles, abilities, and strengths (Andrews, Bonta, & Wormith, 2006).

The risk-needs-responsivity (RNR) model has grown in popularity over the last several decades and is arguably the prevailing paradigm within North American correctional systems today. Because the RNR model places a premium on prioritizing the higher risk offenders for programming, the development and use of valid and reliable risk and needs assessment instruments are often central to contemporary correctional practice. Notwithstanding the growing emphasis placed on assessing risk and need, there has long been an interest in predicting which offenders are more likely to recidivate.

Bonta and Andrews (2007) pointed out, for example, that prior to the 1950s, offender risk assessment was based mostly on subjective, professional judgments made by correctional staff. Despite the existence of more objective, actuarial methods for assessing offender risk since at least the late 1920s (Burgess, 1928), it was not until the 1970s that clinical judgment began to give way to the development of what Bonta and Andrews (2007) referred to as second-generation risk assessment instruments. Developed on the basis of statistical analyses, these actuarial instruments were found to consistently outperform clinical judgment in predicting recidivism (Brennan, Dieterich, & Ehret, 2009). Yet, because second-generation assessment tools relied almost exclusively on static, criminal history factors to make predictions about future criminal behavior, third-generation instruments later attempted to include dynamic factors that would help identify the areas to target for programming. Building on the need to include both static and dynamic factors, fourth-generation risk assessment instruments have been designed to follow offenders from intake to case closure, be administered to offenders on multiple occasions, and better integrate protective factors within the assessment process (Brennan et al., 2009).

In discussing the characteristics of fourth-generation risk assessment tools, Brennan and colleagues (2009) further noted that these instruments are more likely to be developed with advanced statistical modeling techniques than their predecessors. In the late 1920s, Burgess (1928) devised a method for predicting recidivism among parolees. For items associated with recidivism, Burgess (1928) assigned a value of 1 when the risk factor was present and a value of 0 when it was absent. Adding up the points for each item resulted in a total score, with a higher value indicating a greater risk of recidivism. Since that time, a modified Burgess method has been created that scores items with ordinal, rather than binary, response categories according to the strength of an item's association with recidivism. Both the original and modified Burgess methods have been widely used in the development of second-, third-, and even some fourth-generation recidivism risk assessment tools.

Within the last 15 years, regression modeling has been increasingly used to develop prediction tools in the criminal justice field (Brennan & Oliver, 2000; Duwe, 2012; Duwe, 2014a; Duwe & Freske, 2012; Lowenkamp & Whetzel, 2009). In a recent example, Duwe (2014a) used multiple logistic regression to create the Minnesota Screening Tool Assessing Recidivism Risk (MnSTARR). Developed on nearly 12,500 offenders (11,375 males and 1,100 females) released from Minnesota prisons between 2003 and 2006, the MnSTARR is a "multiple-band" instrument that assesses risk separately for male and female offenders for the following types of recidivism: felony reconviction, reconviction of a non-violent offense, and reconviction of a non-sexual violent offense. Depending on whether they have a sexual offending history, male offenders are also assessed for their risk of committing either a first-time or a repeat sex offense. The results from the development of the MnSTARR showed its predictive discrimination was significantly greater than that of the Level of Service Inventory–Revised, a widely used third-generation instrument (Duwe, 2014a).

Most recently, machine learning methods—namely, random forests—have been used to devise recidivism prediction instruments for Philadelphia probationers (Barnes & Hyatt, 2012; Berk, Sherman, Barnes, Kurtz, & Ahlman, 2009). A branch of artificial intelligence, machine learning focuses on the design and development of algorithms that enable systems to learn from empirical data (Korada, Kumar, & Deekshitulu, 2012). In other words, machine learning algorithms are designed to discover systematic patterns in empirical data without being explicitly programmed to find them based on a priori knowledge. Machine learning algorithms first emerged in the 1980s with the creation of artificial neural networks (ANNs). Since the 1990s, other machine learning approaches have been developed, including support vector machines (SVMs), bagging (i.e., bootstrap aggregating), boosting, and random forests.

The prediction instruments using machine learning algorithms (Barnes & Hyatt, 2012; Berk et al., 2009) are notable for a few reasons. First, our understanding of why someone reoffends is far from perfect, and risk prediction models available today to criminal justice professionals still have ample room for improvement in terms of predictive accuracy. If properly applied, the forecasting performance of machine learning algorithms could be at least as good as traditional parametric modeling such as logistic regression and could be better especially when complex, nonlinear relationships exist between predictors and the outcome of interest (Berk & Bleich, 2014). Machine learning algorithms hold promise for improving our practice in the assessment and classification of offender populations.

Second, the process of machine learning assumes that researchers, who have access to relevant data on study subjects, will use such information to predict the outcome of interest without having to worry about implementation details regarding how to tune each variable for optimal performance as a whole. As such, machine learning algorithms can minimize intermediate human interactions in the process of risk assessment, which, in turn, reduces any individual biases and system costs associated with risk assessment. Widely available recidivism risk assessment instruments are often designed to be scored manually and used in virtually any jurisdiction. In contrast, the tools designed by Berk et al. (2009) and Barnes and Hyatt (2012) are fully automated and specific to one jurisdiction (Philadelphia). As evidenced by the California Static Risk Assessment (CSRA) instrument (Turner, Hess, & Jannetta, 2009), fully automated tools can be developed with other methods besides machine learning algorithms. Still, because most machine learning methods can effectively filter out unimportant predictors and handle data sets with large numbers of variables (Hastie, Tibshirani, & Friedman, 2009), these methods appear to be tailor-made to operate within an automated environment. Automating the risk assessment process helps improve the efficiency of correctional systems by removing inter-rater error and eliminating the time correctional staff spend manually scoring assessments.

Third, although the automated instruments noted above were recently created, all three are more like the older second-generation instruments insofar as they consider mostly static, criminal history factors. More specifically, none of the instruments was designed to capture dynamic risk factors or to be administered on multiple occasions to the same offender. To our knowledge, the field has yet to implement a fully automated, fourth-generation recidivism risk assessment instrument based on a machine learning algorithm.

Prior Research on the Predictive Performance of Machine Learning Algorithms

Research from other disciplines has generally shown that the newer machine learning algorithms outperform older techniques such as logistic regression, decision trees, and Naïve Bayes. For instance, Caruana and Niculescu-Mizil (2006) conducted an extensive empirical comparison of commonly used supervised learning algorithms, which attempt to predict the value of an outcome measure based on a number of input measures. Using data sets from different substantive domains, Caruana and Niculescu-Mizil (2006) found there is no universally best supervised learning algorithm, which is consistent with the "no free lunch" theorem stated by Wolpert (1996). That is, even the best models performed poorly sometimes, while models with poor overall performance sometimes performed well. Still, Caruana and Niculescu-Mizil (2006) found there were some methods that generally performed better than others. Boosted trees performed the best overall, followed closely by random forests. The next best group of classifiers consisted of bagged trees, SVMs, and ANNs, while the worst performing group included boosted stumps, Naïve Bayes, decision trees, and logistic regression.

Within corrections, there have been at least six prior studies that, similar to the one by Caruana and Niculescu-Mizil (2006), compared the performance of various methods in predicting recidivism. In the study by Breitenbach, Dieterich, Brennan, and Fan (2009), they compared predictive accuracy among J48, SVM, ANN, random forests, alternating decision trees, and logistic regression. Breitenbach et al. (2009) found random forests did especially well in predicting violent recidivism in the full set of variables they used, whereas more traditional techniques such as logistic regression did better when a subset of predictors was selected.

The study by Liu, Yang, Ramsey, Li, and Cold (2011) compared the predictive performance of logistic regression, ANN, decision trees, and discriminant analysis among 1,657 offenders (1,353 males and 304 females). They found little difference among these methods in predicting violent recidivism. Similarly, in their study examining logistic regression, multivariate adaptive regression splines (MARS), discriminant analysis, decision trees, adaptive boosting, LogitBoost, ANN, SVM, K-nearest neighbors, and partial least squares, Tollenaar and van der Heijden (2013) found the newer machine learning algorithms did no better than standard techniques such as logistic regression in predicting several types of recidivism. In their analysis of CSRA data, Hess and Turner (2013) reported that use of random forests produced a significant, albeit modest, improvement in predictive discrimination over logistic regression. Berk and Bleich (2013) found that random forests had greater accuracy in predicting recidivism than stochastic gradient boosting which, in turn, performed slightly better than logistic regression. Most recently, Hamilton, Neuilly, Lee, and Barnoski (2014) reported that the performance of logistic regression was comparable with random forests and ANN in predicting four types of recidivism.

One reason why some prior studies have not found a difference in predictive performance among the methods they assessed, Berk and Bleich (2013) argued, is that not all of the techniques were properly "tuned." For example, there are some methods, such as ANN, which simply require more parameter adjustments to maximize predictive performance. Although Berk and Bleich (2013) acknowledged the difficulty in achieving a perfect "apples to apples" comparison among different classifier methods, they still suggest that for the forecasting contest to be fair, each method (or contestant) should be tuned for optimal performance.

Given the relative scarcity of evaluations within corrections that have assessed the predictive accuracy of the newer machine learning methods alongside the older, more established techniques, some questions remain as to whether machine learning algorithms offer a superior alternative and, if so, to what extent. Moreover, given that Caruana and Niculescu-Mizil (2006) found there is not a single classifier that performs the best in every situation, it is unclear the types of situations in which some methods will generally perform better than others. For example, although it is important for a risk assessment tool to accurately predict recidivism in general, recent research has demonstrated the importance of being able to accurately predict violent recidivism, which is more serious and costly to society but also less common (Berk et al., 2009; Duwe, 2012). Therefore, identifying methods that are especially accurate in predicting rare events is critical. In addition, not much is known about how well each method performs across various sample sizes or with different numbers of predictor variables available.

Present Study

Due to the limited number of studies within criminal justice that have compared the performance of machine learning and traditional methods, scholars have recently argued that more empirical evaluations are needed (Brennan & Oliver, 2013; Bushway, 2013; Ridgeway, 2013a). Among prior recidivism prediction contests, Tollenaar and van der Heijden (2013) conducted the broadest assessment by examining 11 different classification methods. In this study, we assess the performance of a dozen supervised learning algorithms in predicting recidivism. In particular, we examine simple logistic regression (i.e., a main effects model), full logistic regression (i.e., a model with non-linear and interaction terms), regularized logistic regression, decision trees, Naïve Bayes, ANN, SVM, bagged trees, random forests, LogitBoost, MultiBoosting, and logistic model trees (LMTs). In doing so, we include most of the algorithms evaluated by Caruana and Niculescu-Mizil (2006) as well as several promising newer ones (e.g., MultiBoosting and LMTs). Moreover, we assess the performance of logistic regression by not only including regularized and unregularized models but also estimating models with interaction and nonlinear terms as well as those with only main effects.[1]

Previous recidivism prediction contests have analyzed performance across as many as four different scenarios (Hamilton et al., 2014), which have typically varied by type of recidivism. To gain a better understanding of the potential relative strengths and weaknesses of each algorithm, we compare predictive performance across 10 different scenarios that vary by gender and type of recidivism. We extend existing research by assessing how well each method performs across varying sample sizes, baseline recidivism rates, and the number of predictors available. We evaluate the performance of each algorithm by examining its predictive discrimination, accuracy, and calibration.

The data used in this study are derived, to a large extent, from the data set used to develop the MnSTARR (Duwe, 2014a). The MnSTARR data set is, in several ways, ideally suited to assess performance across a variety of conditions. First, the MnSTARR assesses risk of five different types of recidivism with base rates that range from a high of 47% to a low of less than 1%. Second, because the MnSTARR assesses risk separately for male and female offenders, we examine performance across varying levels of sample sizes. Last, depending on the type of recidivism measure assessed, the number of variables in the data set ranges from a high of 55 to a low of 10.

In the following section, we briefly review the supervised learning algorithms we evaluate in this study. Next, we describe the data and methods used here. Following a presentation of the results, we conclude by discussing the implications of the results for recidivism risk assessment within corrections.

An Overview of Supervised Learning Algorithms

In supervised learning, the goal is to predict the value of an outcome measure based on a number of input measures. In unsupervised learning, however, there is no outcome measure; rather, the goal is to describe the associations and patterns among a set of input measures such as principal components analysis (Hastie et al., 2009). Below, we provide a brief overview of the 12 supervised learning methods considered in this study.

Logistic Regression

Developed in the 1940s, logistic regression has been a popular statistical method for creating prediction instruments over the last several decades, particularly in the biomedical field. Because logistic regression can be sensitive to irrelevant predictors, stepwise selection procedures are typically used to identify the relevant (i.e., statistically significant) variables in the model. Nevertheless, there are regularization methods, such as ridge regression, that can effectively handle data sets with a relatively large number of predictors—and the accompanying problems with collinearity—by shrinking overly large parameter estimates. In doing so, regularized logistic regression also helps mitigate overfitting.

Logistic regression offers a relatively high degree of transparency because the prediction model is expressed in a formula that includes only the relevant variables used to calculate the predicted outcome. It is possible to not only discern the extent to which each variable contributes to the prediction but also calculate the predicted recidivism probability for a specific offender. However, unlike the other methods discussed below, logistic regression does not automatically account for interactions and nonlinear relationships. Instead, any interactions and nonlinear effects must be performed manually and identified prior to estimating the model. Nevertheless, there are some linear regression methods (e.g., splines, fractional polynomials, generalized additive models, or kernel logistic regression) where nonlinearity can be estimated automatically.

Naïve Bayes

Based on the application of Bayes's theorem, Naïve Bayes is a simple probabilistic classifier that assumes the presence or absence of a particular feature of a class is unrelated to the presence or absence of any other feature. Due to this assumption, only the variances of the predictors are required for prediction purposes. Furthermore, Naïve Bayes is a relatively fast, efficient statistical learning method that needs only a small amount of training data for classification (Hastie et al., 2009). Naïve Bayes models numeric attributes through a normal distribution, a kernel density estimator, or supervised discretization. Although it is has been found to be an effective classifier despite its simplicity, Naïve Bayes has been shown to yield poorly calibrated probabilities, with values pushed close to 0 and 1 (Bennett, 2000).

Decision Trees

Classification and regression trees (CART) were created in the 1980s by Breiman, Friedman, Olshen, and Stone (1984). In predicting recidivism, which is typically measured as a binary outcome, classification trees (i.e., decision trees) are used. The construction of a tree begins by splitting the data into smaller nodes based on a splitting criterion. The decision tree "decides" whether to continue splitting the data or declare a node terminal. Each terminal node is assumed to have a homogeneous distribution. For classification, the predicted class is the most common class in the node (majority vote). It is also possible, however, to generate an estimated probability of membership in each class (recidivist or non-recidivist). If the final tree is too big, however, it can be pruned to reduce its size so as to increase its classification performance. Although sensitive to small changes in the training data, decision trees automatically select variables, are relatively easy to interpret, and can handle interaction and nonlinear effects (Hastie et al., 2009).

ANN

At around the same time Breiman and colleagues created CART, ANNs began to emerge. Developed as models for the human brain, ANNs encompass a large class of models and learning methods (Hastie et al., 2009). ANNs are essentially nonlinear statistical models that take nonlinear functions of linear combinations of the variables in the model. The units in the middle of the network are known as hidden units because the values are not directly observed. In general, the more hidden units an ANN has, the more nonlinear it is. ANNs learn the data through backpropagation, wherein each hidden unit passes and receives information only to, and from, units that share a connection. The learning process, moreover, is iterative in that ANNs can take multiple sweeps, referred to as epochs, through the data set. ANN has unknown parameters, called weights, and values are identified in an effort to make the model fit the training data well. ANN minimizes overfitting,[2] however, through weight decay, which is used to penalize the weight in each iteration. Because it is difficult to determine how specific classification decisions are made, ANNs are relatively opaque. Nevertheless, ANNs have been found to be particularly effective where prediction without interpretation is needed (Hastie et al., 2009).

SVMs

The 1990s saw the emergence of a number of new machine learning algorithms. One of these was SVM, which was developed by Vapnik (1995). SVM attempts to create an optimal separating hyperplane between two classes (e.g., recidivists and nonrecidivists). Support vectors are data points that lie closest to the decision surface and are the most difficult to classify. If removed, they would change the position of the dividing hyperplane. The decision function is therefore fully specified by the support vectors. SVM attempts to maximize the margin around the separating hyperplane. For nonlinear classification, SVM produces nonlinear boundaries by constructing a linear boundary in a large, transformed version of the feature space. SVM uses a kernel function to extend patterns that are not linearly separable by transformations of original data to map into new space. SVM has been found to be an effective classifier that is not susceptible to overfitting or noisy data (Hastie et al., 2009).

Bootstrap Aggregating (Bagging)

Building on his research on CART, Breiman (1996) developed bagging in the 1990s. An early ensemble method, bagging produces multiple versions of a predictor, which are then used to create an aggregated predictor. Bagging constructs multiple decision trees by repeatedly resampling training data with replacement and voting the trees for a consensus prediction. Although bagging has little impact on bias, it has been shown to reduce the variance of the classifier. Furthermore, as Breiman (1996) pointed out, bagging produces a complex model that is not as transparent as, say, a single decision tree, but it nevertheless yields more accurate predictions.

Random Forests

Breiman (2001) further extended his work by later creating random forests. Similar to bagging, random forests is an ensemble method. Random forests involve growing a forest of many trees, and each tree is grown on an independent bootstrap sample from the training data. Unlike bagging, however, each time a tree is fit at each node, some of the predictor variables are censored. Random forests then find the best split based on the selected predictor variables. The trees are grown to a maximum depth, and a consensus prediction is obtained after voting the trees. Like bagging, random forests generally produce complex models that achieve relatively high predictive accuracy. Random forests also provide information on the importance of each variable for the classification model.

Boosting

Created by Freund and Schapire (1996) during the 1990s, boosting is another ensemble method that tries to increase predictive accuracy by forming a powerful committee. Boosting attempts to improve classification by identifying weak learners with low correlation and bias and then combines the output of these learners to yield a prediction. As Berk and Bleich (2013) explained, weak learners are "boosted" to perform as strong learners through an iterative reweighting of the data whereby more weight is given to observations that are more difficult to classify. Although adaptive boosting (i.e., AdaBoost) was the first boosting algorithm (Freund & Schapire, 1996), others have since been developed. Friedman, Hastie, and Tibshirani (2000) created LogitBoost, which we examine here. We also evaluate MultiBoosting, a method developed by Webb (2000) that combines boosting and wagging, a variant of bagging. Wagging requires a base learning algorithm that uses training cases with differing weights. Rather than using random bootstrap samples to form the successive training sets, it assigns random weights to the cases in each training set (Webb, 2000).

LMTs

The most recent method we assess in this study is LMTs, which was developed by Landwehr, Hall, and Frank (2005). LMT combines tree induction methods and logistic regression models. The trees contain linear regression functions at the leaves. A stagewise fitting process is used to construct logistic regression models at the leaves by using boosting to incrementally refine those constructed at higher levels in the tree. To minimize classification error, LMT uses the same pruning method as that found within CART (Landwehr et al., 2005).

Data and Method

The original MnSTARR data set consisted of 12,475 individual offenders released from Minnesota prisons between 2003 and 2006. For this study, we expanded the release window by 4 years so that the data include offenders released between 2003 and 2010. When offenders had multiple releases during the 2003-2010 period, we used the offender's first release from prison. The total sample size consists of 27,772 individual offenders. Of these, 24,917 are male offenders, while the rest (2,805) are female offenders. To approximate as much as possible the actual conditions under which risk assessment instruments are used, we selected offenders released in the two most recent years (2009 and 2010) as our test set (5,888 males and 789 females). The predictive validity results reported below are based on each algorithm's performance on the test set. Our training set, which we used to develop the models, consisted of offenders released from prison between 2003 and 2008 (19,029 males and 2,016 females).

The sample sizes vary for the two sexual offending measures for male offenders. No sexual offending measures were developed for females because none of the 2,805 offenders recidivated with a sex offense conviction. Of the 24,917 male offenders, 20,420 did not have a prior sex offense conviction in their history. For the first-time sexual offending measure, 15,554 of these offenders were placed in the training set because they were released from prison prior to 2009. The remaining 4,866 offenders were placed in the test set because they were released from prison in 2009 and 2010.

The remaining 4,497 male offenders had a prior sex offense conviction in their criminal history and, thus, were assessed for their risk of sex offense recidivism. The Minnesota Sex Offender Screening Tool–3 (MnSOST-3), which is integrated within the MnSTARR, is the instrument used to assess sexual recidivism risk for Minnesota sex offenders. Part of the development sample for the MnSOST-3 consisted of 220 sex offenders released from prison during the early 1990s (Duwe & Freske, 2012). Although only a limited number of variables (10) were available for these offenders, they were included in the training set for sexual recidivism so as to provide a scenario with a smaller number of predictors. The training set for sexual recidivism therefore contains 3,695 sex offenders released from prison prior to 2009, while the test set holds 1,022 sex offenders released from prison during 2009 and 2010.

In Table 1, we present basic descriptive statistics on the predictors in the data set we used. More specifically, we present the mean and standard deviations of the predictors used in the training and test sets for both male and female offenders. Most of the predictors we analyzed are static factors, and a majority of these factors are measures of criminal history. The dynamic factors, however, are variables whose values could change while an offender was incarcerated. The list of dynamic factors covers institutional behavior that can either elevate (e.g., discipline convictions or identification as an active gang member) or mitigate (e.g., completion of chemical dependency treatment or receiving a visit in prison) an offender's recidivism risk.

Table 1. Descriptive Statistics for MnSTARR Predictors.

		Male offenders				Female Offenders			
		Training		Test		Training		Test	
Predictors	Description	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Static/criminal history									
Prior supervision failures	Total number of revocations on probation and parole	1.190	1.367	0.780	0.952	1.150	1.067	0.840	0.761
Total convictions	Total number of convictions (any offense level)	8.950	8.073	8.400	6.565	9.050	8.881	8.140	7.559
Felony convictions	Total number of felony convictions	4.150	3.657	3.410	2.845	3.500	3.472	2.690	2.345
Felony specialization/ diversity	Degree of specialization/diversity in felony offenses	0.594	0.354	0.651	0.350	0.674	0.332	0.670	0.364
Violent convictions	Total number of violent offense convictions	1.520	1.842	1.450	1.753	0.660	1.235	0.580	1.063
Violent specialization/ diversity	Degree of specialization/diversity in violent offenses	0.879	0.241	0.899	0.209	0.951	0.164	0.935	0.205
Total assault convictions	Total number of assault offense convictions	1.140	1.685	1.230	1.652	0.470	1.010	0.420	0.917
Robbery convictions in last 5 years	Total number of robbery convictions in last 5 years	0.130	0.513	0.110	0.551	0.050	0.273	0.040	0.297
VOFP convictions	Total VOFP, stalking, and harassment convictions	0.220	0.713	0.250	0.744	0.070	0.378	0.070	0.364
Disorderly conduct convictions	Total number of disorderly conduct convictions	0.420	0.996	0.510	0.971	0.290	0.738	0.330	0.771
Prostitution convictions	Total number of prostitution offense convictions	0.010	0.138	0.010	0.186	0.370	1.681	0.150	0.958

(continued)

Table 1. (continued)

		Male offenders				Female Offenders			
		Training		Test		Training		Test	
Predictors	Description	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Drug offense convictions	Total number of drug offense convictions	1.020	1.350	0.870	1.202	1.200	1.524	0.970	1.212
Drug offense specialization/diversity	Degree of specialization/diversity in drug offenses	0.900	0.234	0.923	0.203	0.866	0.265	0.866	0.282
False information to police cons.	Total number of false information to police convictions	0.260	0.692	0.270	0.700	0.350	0.864	0.360	0.847
Flee/Escape police convictions	Total number of flee/escape police convictions	0.230	0.603	0.240	0.587	0.100	0.359	0.110	0.342
Weapons offense convictions	Total number of weapons offense convictions	0.130	0.375	0.140	0.407	0.020	0.154	0.010	0.146
Burglary convictions	Total number of burglary convictions	0.650	1.515	0.510	1.100	0.140	0.605	0.140	0.476
Theft convictions	Total number of theft convictions	1.530	2.981	1.300	2.116	2.250	4.334	1.850	2.936
Fraud/Forgery convictions	Total number of fraud/forgery convictions	0.500	1.525	0.380	1.216	1.550	2.728	1.220	2.470
Other property convictions	Total number of other property convictions	0.460	0.901	0.450	0.890	0.270	0.666	0.270	0.697
Total property convictions	Total number of property offense convictions	3.140	4.834	2.640	3.618	4.220	6.168	3.490	5.027
Property offense specialization/diversity	Degree of specialization/diversity in prop. offenses	0.838	0.250	0.842	0.249	0.724	0.333	0.747	0.339
DWI offense convictions	Total number of DWI Offense Convictions	0.600	1.384	0.810	1.587	0.520	1.159	0.630	1.279

(continued)

Table 1. (continued)

		Male offenders				Female Offenders			
		Training		Test		Training		Test	
Predictors	Description	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Any juvenile convictions	Convicted or adjudicated delinquent for an offense	0.290	0.452	0.340	0.475	0.100	0.298	0.100	0.302
4th-5th degree assaults in last 3 years	4th and 5th degree assault convictions in last 3 years	0.120	0.411	0.130	0.430	0.050	0.261	0.080	0.333
Robbery convictions <21	Total number of robbery convictions under age of 21	0.080	0.357	0.090	0.361	0.020	0.147	0.020	0.166
Repeat sex offending specific									
Sex offender	Sex offender = 1; non-sex offender = 0	0.180	0.386	0.170	0.379				
Predatory offenses	Total number of predatory offense convictions	0.250	0.661	0.200	0.554				
Predatory off. specialization/diversity	Degree of specialization/diversity in pred. offenses	0.946	0.266	0.960	0.174				
Male victims	Number of predatory offenses involving male victims	0.010	0.142	0.010	0.121				
Stranger victim	At least one sex offense involving stranger	0.030	0.180	0.030	0.173				
Public place	Sex offense committed in a public location	0.020	0.152	0.020	0.146				
Failure to register convictions	Total number of failure to register convictions	0.050	0.280	0.050	0.274				
Disorderly conduct cons. last 3 yrs.	Number of disorderly conduct convictions in last 3 years	0.020	0.145	0.120	0.426				

(continued)

Table 1. (continued)

		Male offenders				Female Offenders			
		Training		Test		Training		Test	
Predictors	Description	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Complete sex offender/CD treatment	Completed sex offender/CD treatment in prison	0.020	0.123	0.010	0.092				
Intake									
New court commitment	Admitted to prison directly from court	0.510	0.500	0.600	0.491	0.370	0.484	0.390	0.489
Probation violator	Admitted to prison for probation violation	0.350	0.475	0.380	0.485	0.440	0.496	0.570	0.496
Release violator	Admitted to prison for parole violation	0.150	0.354	0.030	0.158	0.190	0.391	0.040	0.194
Metro county commit	Admitted to prison from twin cities metro area	0.500	0.500	0.490	0.500	0.480	0.500	0.390	0.487
Marital status	Married = 1; Not married = 0	0.120	0.325	0.120	0.331	0.100	0.305	0.110	0.315
Dynamic									
Length of stay in prison (months)	Difference in months between admission and release	18.185	22.151	22.591	29.571	10.280	14.201	14.731	23.261
Suicidal history	History of suicidal tendencies	0.090	0.288	0.120	0.329	0.200	0.402	0.290	0.452
Prison discipline convictions	Total number of discipline convictions during current term	2.700	6.541	4.140	10.924	3.930	10.208	7.350	15.450
STG	Active member of STG	0.070	0.255	0.030	0.173	0.010	0.083	0.000	0.036
Ever visited in prison	Visited at least once during current term	0.510	0.500	0.560	0.496	0.460	0.499	0.550	0.497
Secondary degree at release	Offender has GED or high school diploma	0.750	0.434	0.770	0.418	0.680	0.465	0.750	0.435

(continued)

Table 1. (continued)

		Male offenders				Female Offenders			
		Training		Test		Training		Test	
Predictors	Description	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Post-secondary degree at release	Offender has post-secondary degree/ certificate	0.060	0.233	0.080	0.277	0.060	0.235	0.090	0.286
Complete CD treatment	Completed chemical dependency treatment in prison	0.130	0.339	0.180	0.381	0.150	0.355	0.130	0.339
Complete IFI (faith-based program)	Complete InnerChange Freedom Initiative	0.020	0.122	0.020	0.150	0.000	0.000	0.000	0.000
Complete EMPLOY (employment program)	Complete EMPLOY	0.010	0.098	0.050	0.218	0.020	0.156	0.090	0.292
Complete cognitive-behavioral program	Completed cognitive-behavioral programming	0.040	0.199	0.090	0.287	0.140	0.342	0.250	0.431
Entered work release	Entered work release	0.120	0.319	0.130	0.332	0.160	0.363	0.170	0.375
Entered CIP (boot camp)	Entered challenge incarceration program	0.050	0.213	0.070	0.254	0.080	0.277	0.100	0.293
Age at release	Age in years at time of release	33.380	9.931	33.600	10.687	34.790	8.901	34.320	9.603
Unsupervised release	Released to no correctional supervision	0.080	0.272	0.050	0.211	0.130	0.338	0.060	0.234
N		19,029		5,888		2,016		789	

Note. MnSTARR = Minnesota Screening Tool Assessing Recidivism Risk; STG = Security Threat Group; GED = general education development; IFI = InnerChange Freedom Initiative; CIP = Challenge Incarceration Program; CD = chemical dependency; VOFP = violation of order for protection; DWI = driving while intoxicated.

Supervised Learning Algorithm Parameters

We used several statistical and data mining software packages to carry out the analyses for this study. In developing the main effects and nonlinear/interaction term logistic regression models, we relied on Stata. We also used Stata to estimate Brier scores to assess calibration. For the remaining 10 models, we used the Konstanz Information Miner (KNIME), which is integrated with packages such as R and Waikato Environment for Knowledge Analysis (WEKA). In particular, we relied mainly on the WEKA extensions within KNIME to generate these models.

Several types of validation methods are available to determine the reproducibility of a prediction model. The split-population, or data splitting, method involves using a portion (e.g., one half or two thirds) of the sample to develop the prediction tool (the training set), which is then applied to the remaining portion of the sample to test the internal validity of the model (test set). Conversely, cross-validation, or k-fold validation, is more efficient than the split-population method because it involves repeated data splitting, whereas research has shown that bootstrap resampling is the most efficient validation technique (Steyerberg, Bleeker, Moll, Grobbee, & Moons, 2003; Steyerberg et al., 2001). As with most of the aforementioned prediction contests (see Hamilton et al., 2014, for an exception), we relied on data splitting as our validation method. Although it is the least efficient method, the size of the training and test sets we used was still relatively large. In using the data splitting approach, we varied the parameters for each algorithm on the training set to identify the best model and then assessed predictive performance on the test set.

For the simple logistic regression models, which did not include interaction or nonlinear effects, we used backward stepwise selection and bootstrap resampling to select predictors (Duwe, 2014a). For full logistic regression, we entered interaction and quadratic terms in the models. We varied the ridge estimator value (1.0E-8 to 500) in the regularized logistic regression models. For the decision tree models, we varied the quality measure (Gini index or Gain ratio), pruning method (no pruning or minimum description length (MDL)), presence/absence of reduced error pruning, minimum number of observations per node (2-500), and number of threads (1-5) parameters. We varied the parameters for Naïve Bayes by modeling them as a single normal, using a kernel estimator or discretizing them using supervised discretization.

With the SVM models, we varied the kernel (linear or polynomial), cost (0-1), gamma (0-1), and epsilon (0-1) parameters. For ANN, we used Multilayer Perceptron and varied the number of hidden layers (1-10), the learning rate (0-1), momentum (0-1), epochs (5-1,000), and the presence/absence of decay. For random forests, we varied the depth of the trees (0-20), the number of trees (2-1,000), the number of slots (1-5), and the number of features (0-15). With bagging, we used the default classifier (reduced error pruning (REP) tree) because it produced the best results and varied the parameters by the number of execution slots (1-5), number of iterations (2-1,000), and bag size percent (70-100). We also used the default classifier (decision stumps) for both LogitBoost and MultiBoosting because they produced the best results. For LogitBoost, we varied the number of folds (0-2),[3] iterations (2-1,000), runs (1-5), shrinkage (0.05-3.00), presence/absence of resampling, and weight threshold (70-100). With MultiBoosting, we varied the number of iterations (2-1,000), subcommittees (1-10), presence/absence of resampling, and weight threshold (70-100). For LMTs, we varied the errors on probabilities (true or false), maximum boosting iterations (2-1,000), heuristic stops (2-1,000), use of Akaike Information Criterion (AIC; true or false), use of cross-validation (true or false), and beta weight (0.0-0.5).

Predictive Performance Measures

As noted above, we evaluated performance by analyzing predictive discrimination, accuracy, and calibration. Predictive discrimination examines how well the model separates the recidivists from the non-recidivists. The area under the curve (AUC) is a widely used statistic to estimate the predictive discrimination of risk assessment instruments. With values that range from 0 to 1, the AUC is interpreted as the probability that a randomly selected recidivist has a higher score on a risk assessment instrument than a randomly selected non-recidivist. Values at either end of the spectrum (0 or 1) reflect complete separation between recidivists and non-recidivists, whereas a value of 0.50 indicates the prediction tool does no better than chance. Although the AUC uses different misclassification cost distributions for different classifiers and can provide misleading results if receiver operating characteristic (ROC) curves cross (Hand, 2009), it is relatively independent of base rates and selection ratios across studies (Smith, 1996).

To measure predictive accuracy, we used three threshold-based metrics—accuracy (ACC), precision, and recall—that are, unlike the AUC, more sensitive to different base rates, particularly those for low-frequency recidivism events that approach zero. ACC assesses the extent to which each algorithm correctly classified offenders as recidivists or non-recidivists. For example, if a recidivist had a predicted recidivism probability less than 50%, then this offender would have been incorrectly classified as a non-recidivist (i.e., false negative). Conversely, if this offender had not recidivated, then she or he would have been accurately classified (i.e., true negative).

Precision measures the percent of positive predictions that were correct (based on the 50% threshold), whereas recall reflects the percentage of positives (i.e., recidivists) that were captured. To illustrate, let us assume we were predicting recidivism for a sample of 100 offenders. Furthermore, we predicted 45 would be recidivists (i.e., predicted probability is equal to or greater than 50%), whereas 50 actually recidivated. Of the 45 positive predictions, 30 were recidivists, resulting in 0.67 for precision. Given that there were 50 recidivists overall, the 30 positives we captured would yield a recall value of 0.60.

Along with predictive discrimination and accuracy, calibration is an important metric that should be considered when assessing the performance of supervised learning algorithms. Metrics such as the AUC address relative risk by assessing how accurately an instrument is able to identify offenders who are more, or less, likely to recidivate in comparison with a broader population of offenders. Within corrections, relative assessments of risk are often all that is needed because the assessments are mainly used to help improve the allocation of resources by prioritizing higher risk offenders for interventions (i.e., the risk principle). There are some instances, however, where it is also necessary to assess an offender's absolute risk of recidivism. Perhaps the most notable example involves the use of risk assessment instruments to help determine whether a convicted sex offender who has served his prison sentence should be involuntarily civilly committed. One of the common criteria for commitment in the 20 states that operate sex offender civil commitment programs is that the offender's likelihood of recidivism over the rest of their lifetime is "more likely than not," which generally translates to a probability of greater than 50% (Duwe, 2014b).

For a prediction instrument to make accurate absolute assessments of risk, the model's predicted probabilities must be calibrated with the observed recidivism outcomes.[4] For example, let us assume we have 100 offenders who have recidivism probabilities between 45% and 55%, which amounts to an average of 50%. If our model is well-calibrated, we would expect to see 50 recidivists out of the 100 offenders (i.e., the recidivism rate of 50% equals the average of the predicted probabilities). If, however, the observed recidivism rate was, say, 30% for the 100 offenders, then it would be a poorly calibrated model that overestimates absolute risk. Conversely, if the observed recidivism rate was 70%, then the model would underestimate absolute risk.

Lowess plots are commonly used to visually inspect calibration. Although the Hosmer–Lemeshow test is often used to evaluate calibration, the main disadvantage with this statistic is that it is very sensitive to sample size. In contrast, because the Brier score, a quadratic scoring rule that calculates the squared differences between observed and predicted values, is less sensitive to sample size, it is the more appropriate calibration statistic for larger samples.

Of the algorithms we evaluated, three—ANN, Naïve Bayes, and MultiBoosting—initially yielded poorly calibrated models for all 10 scenarios. More specifically, we estimated Brier scores, and the Spiegelhalter *z* statistic was statistically significant in each scenario for these three methods. To address poor calibration for these algorithms, we utilized stacking, which combines several classifiers. Our base classifier for stacking consisted of one of these three algorithms, while we used logistic regression as the meta-classifier. This approach significantly improved the calibration in most, but not all, of the scenarios.

Results

In Table 2, we present the predictive performance results for the 120 different scenarios (12 algorithms × 10 scenarios) we examined. Table 2 shows each method's AUC, precision, recall, and ACC for the 10 scenarios, and the averages are provided for male offenders, female offenders, and overall. For each scenario, we also denote the models that produced poorly calibrated models with an asterisk (*). The far right-hand column of Table 2 provides the ranking of each algorithm (relative to the other 11) for the predictive performance metrics, and the average overall ranking has been bolded for each method. Last, Table 2 provides information on the training and test set sizes, the recidivism base rates, and the number of predictors available for each scenario.

Table 2. Supervised Learning Algorithm Predictive Performance by Gender and Recidivism Type.

	Males						Females				Overall average and rank			
	GR	FR	NVR	NSVR	FTSO	RSO	GR	FR	NVR	NSVR	Male	Female	Total	Rank
Simple logit														**8.50**
AUC	0.786	0.773	0.762	0.786	0.746	0.707	0.761	0.744	0.738	0.704	0.760	0.737	0.751	11
Precision	0.711	0.709	0.696	0.549	0.000	0.000	0.716	0.667	0.706	0.000	0.444	0.522	0.475	10
Recall	0.623	0.314	0.454	0.115	0.000	0.000	0.400	0.192	0.298	0.000	0.251	0.223	0.240	7
ACC	0.718	0.754	0.719	0.854	0.995	0.984	0.697	0.808	0.705	0.929	0.837	0.785	0.816	6
Full logit														**6.00**
AUC	0.791	0.779	0.767	0.785	0.743	0.745	0.741	0.743	0.747	0.735	0.768	0.742	0.758	9
Precision	0.703	0.673	0.668	0.553	0.000	0.000	0.657	0.638	0.663	0.286	0.433	0.561	0.484	9
Recall	0.651	0.407	0.510	0.161	0.000	0.000	0.479	0.222	0.404	0.038	0.288	0.286	0.287	1
ACC	0.720	0.762	0.719	0.855	0.995	0.984	0.692	0.809	0.714	0.929	0.839	0.786	0.818	5
Regularized logit														**8.25**
AUC	0.787	0.775	0.764	0.790	0.772	0.707	0.764	0.753	0.761	0.766	0.766	0.761	0.764	6
Precision	0.712	0.710	0.700	0.544	0.000	0.000	0.827	0.818	0.737	0.000	0.444	0.596	0.505	5
Recall	0.623	0.315	0.441	0.099	0.000	0.000	0.289	0.054	0.099	0.000	0.246	0.111	0.192	12
ACC	0.718	0.754	0.717	0.853	0.995	0.984	0.692	0.797	0.669	0.933	0.837	0.773	0.811	10
Decision trees														**9.75**
AUC	0.764	0.758	0.751	0.745	0.669	0.692	0.719	0.727	0.747	0.762	0.730	0.739	0.733	12
Precision	0.681	0.649	0.659	0.460	0.000	0.000	0.706	0.462	0.722	0.000	0.434	0.473	0.434	11
Recall	0.606	0.351	0.510	0.118	0.000	0.000	0.397	0.108	0.340	0.000	0.243	0.211	0.243	5
ACC	0.696	0.743	0.715	0.847	0.995	0.979	0.693	0.785	0.717	0.933	0.829	0.782	0.810	11
Support VM														**7.50**
AUC	0.788	0.775	0.762	0.735	0.719	0.701	0.788	0.781	0.775	0.782	0.747	0.782	0.761	7
Precision	0.720	0.707	0.701	0.481	0.000	0.000	0.836	0.700	0.813	0.000	0.435	0.587	0.496	6
Recall	0.617	0.306	0.401	0.058	0.000	0.000	0.324	0.084	0.248	0.000	0.230	0.164	0.204	11

(continued)

Table 2. (continued)

	Males						Females				Overall average and rank			
	GR	FR	NVR	NSVR	FTSO	RSO	GR	FR	NVR	NSVR	Male	Female	Total	Rank
ACC	0.720	0.753	0.711	0.851	0.995	0.984	0.705	0.798	0.711	0.933	0.836	0.787	0.816	6
Neural network														**5.75**
AUC	0.788	0.776	0.763	0.783	0.756	0.726	0.756	0.751	0.754	0.759	0.765	0.755	0.761	7
Precision	0.707	0.680	0.693	0.510	0.000	0.000	0.720	0.594	0.665	0.308	0.432	0.572	0.488	8
Recall	0.573	0.392	0.424	0.175	0.000	0.000	0.343	0.341	0.429	0.075	0.261	0.297	0.275	2
ACC	0.719	0.761	0.711	0.852	0.995	0.984	0.684	0.811	0.719	0.926	0.837	0.785	0.816	6
Naïve Bayes														**9.00**
AUC	0.763	0.753	0.743	0.769	0.792	0.731	0.752	0.727	0.734	0.763	0.759	0.744	0.753	10
Precision	0.727	0.553	0.588	0.000	0.000	0.000	0.770	0.659	0.727	0.000	0.311	0.539	0.402	12
Recall	0.674	0.521	0.592	0.000	0.000	0.000	0.394	0.162	0.312	0.000	0.298	0.217	0.266	3
ACC	0.692	0.729	0.689	0.853	0.995	0.984	0.711	0.805	0.712	0.933	0.824	0.790	0.810	11
Bagged trees														**4.75**
AUC	0.791	0.786	0.769	0.779	0.783	0.688	0.786	0.753	0.790	0.773	0.766	0.776	0.770	4
Precision	0.721	0.717	0.695	0.584	0.000	0.000	0.791	0.632	0.808	0.000	0.453	0.558	0.495	7
Recall	0.619	0.381	0.472	0.102	0.000	0.000	0.397	0.144	0.298	0.000	0.262	0.210	0.241	6
ACC	0.719	0.768	0.722	0.855	0.995	0.984	0.717	0.801	0.724	0.933	0.841	0.794	0.822	2
Random forests														**4.50**
AUC	0.792	0.784	0.769	0.789	0.814	0.746	0.789	0.759	0.784	0.780	0.782	0.778	0.781	1
Precision	0.714	0.703	0.682	1.000	0.000	0.000	0.832	0.600	0.839	0.000	0.517	0.568	0.537	4
Recall	0.634	0.355	0.454	0.003	0.000	0.000	0.378	0.090	0.259	0.000	0.241	0.182	0.217	10
ACC	0.722	0.760	0.713	0.851	0.995	0.982	0.721	0.795	0.717	0.933	0.837	0.792	0.819	3
LogitBoost														**2.00**
AUC	0.793	0.788	0.770	0.790	0.779	0.746	0.781	0.773	0.788	0.764	0.778	0.777	0.777	2

(continued)

Table 2. (continued)

	Males						Females				Overall average and rank			
	GR	FR	NVR	NSVR	FTSO	RSO	GR	FR	NVR	NSVR	Male	Female	Total	Rank
Precision	0.717	0.718	0.693	0.568	0.000	0.000	0.801	0.698	0.832	0.417	0.449	0.687	0.544	1
Recall	0.640	0.370	0.480	0.119	0.000	0.000	0.359	0.222	0.298	0.094	0.268	0.243	0.258	4
ACC	0.726	0.766	0.723	0.855	0.995	0.984	0.708	0.815	0.728	0.930	0.842	0.795	0.823	1
MultiBoosting														**4.50**
AUC	0.791	0.786*	0.770	0.785*	0.771	0.768	0.781	0.763	0.783	0.775	0.775	0.776	0.775	3
Precision	0.714	0.701	0.688	0.586	0.000	0.000	0.800	0.629	0.810	0.500	0.448	0.685	0.543	2
Recall	0.638	0.380	0.486	0.101	0.000	0.000	0.317	0.132	0.227	0.038	0.268	0.179	0.232	8
ACC	0.723	0.764	0.725	0.855	0.995	0.984	0.696	0.800	0.705	0.933	0.841	0.784	0.818	5
Log. model trees														**5.00**
AUC	0.789	0.779	0.764	0.787	0.764	0.717	0.769	0.758	0.763	0.767	0.767	0.764	0.766	5
Precision	0.732	0.722	0.700	0.535	0.000	0.000	0.787	0.692	0.796	0.429	0.448	0.676	0.539	3
Recall	0.585	0.306	0.445	0.114	0.000	0.000	0.410	0.108	0.277	0.057	0.242	0.213	0.230	9
ACC	0.718	0.755	0.718	0.853	0.995	0.984	0.720	0.801	0.716	0.932	0.837	0.792	0.819	3
Training *n*	19,029	19,029	19,029	19,029	15,554	3,695	2,016	2,016	2,016	2,016				
Training rate	0.471	0.317	0.402	0.144	0.009	0.039	0.392	0.232	0.345	0.089				
Test *n*	5,888	5,888	5,888	5,888	4,866	1,022	789	789	789	789				
Test rate	0.448	0.301	0.378	0.149	0.005	0.016	0.399	0.212	0.357	0.067				
Predictors	55	55	55	55	46	10	49	49	49	49				

Note. GR = general recidivism; FR = felony recidivism; NVR = non-violent recidivism; NSVR = non-sexual violent recidivism; FTSO = first-time sexual offending; RSO = repeat sexual offending; AUC = area under the curve; ACC = accuracy.

*Spiegelhalter z score < .05 (the model's predicted probabilities were poorly calibrated with observed recidivism).

LogitBoost had the best overall average ranking (2.00), as it had the highest total average for precision (0.544) and ACC (0.823), the second-highest AUC (0.777), and the fourth-highest recall rate (0.258). Both Random forests and MultiBoosting had the second-highest overall average ranking (4.50). With an average AUC of 0.781, random forests achieved the best predictive discrimination across the 10 scenarios. MultiBoosting had the second-highest average for precision (0.543) and the third-highest AUC (0.775). Both random forests and MultiBoosting had relatively low rankings—10th and 8th, respectively—for recall.

Bagged trees, LMTs, and ANNs comprised the next best group of performers. With an average of 4.75, bagged trees had the fourth-highest overall ranking. Most notably, bagged trees had the second-highest ACC average (0.822), which means that 82.2% of its individual-level predictions were accurate (given a cutoff score of 0.5) across the 10 scenarios. LMT had the fifth-highest ranking (5.00), and it had the third-highest averages for both precision (0.539) and ACC (0.819). With the sixth-highest overall ranking (5.75), ANN had the highest average recall percentage (0.275) among the algorithms.

SVM, regularized logistic regression, and full logistic regression achieved similar levels of predictive performance overall. Although it ranked 11th for recall, SVM had the 6th-highest averages for precision (0.496) and ACC (0.816). Similarly, regularized logistic regression ranked last for recall, but it had the 5th-highest average for precision (0.505) and 6th-highest average for AUC (0.764). At 7.50, full logistic regression had the 8th-highest overall ranking. It had the 5th-highest ACC.

The worst performing group contained three of the older methods we examined—Naïve Bayes, simple logistic regression, and decision trees. When we focus on average recall rates, each of these methods performed relatively well (i.e., rankings ranged between third and seventh). None of these methods, however, demonstrated strong performance based on the other metrics we assessed. For example, decision trees had the lowest AUC (0.733), whereas Naïve Bayes had the lowest average rate for precision (0.402).

Discussion

Based on a variety of risk and protective factors of recidivism, this study compares the performance of 12 different classification methods applied to offenders released from Minnesota prisons. The findings reported above add to the emerging debate around current policy and practice in risk assessment (Berk & Bleich, 2013; Harcourt, 2006). First, similar to Caruana and Niculescu-Mizil (2006) and Liu et al. (2011), we found that one approach does not work the best in every circumstance. There was even one scenario in which one of the worst classifiers performed better than one of the best classifiers (e.g., Naïve Bayes had a higher AUC than LogitBoost for male first-time sexual offending). On the whole, however, LogitBoost had the strongest overall performance, followed by Random forests, MultiBoosting, bagged trees, and LMTs.

Second, although there was not a wide gulf in predictive performance between the best and worst classifiers, the newer machine learning algorithms generally outperformed the older methods in predicting recidivism. Similar to what Hess and Turner (2013) reported in comparing the performance of logistic regression and random forests, the difference in the average AUC was modest ($\Delta = 0.048$) between the best (0.781 for random forests) and worst (0.733 for decision trees) classifiers. This point is succinctly illustrated in Figure 1.

The graph on the left panel compares two ROC curves for general recidivism among female offenders: LogitBoost—one of the newer machine learning algorithms—and simple logistic regression. Although the visual difference between the two classifiers is marginal, LogitBoost (AUC = 0.781) outperformed simple logistic regression (AUC = 0.761) at the .05 level ($\chi^2 = 4.51$, $df = 1$). Furthermore, the Precision-Recall (PR) curves shown in the right panel indicate that LogitBoost has a notable advantage over simple logistic regression. Because the number of non-recidivists exceeds the number of recidivists in the data, it becomes easy to correctly predict non-recidivists. Consequently, the resulting large number of true negatives could produce a high AUC statistic even when the algorithm does not perform well. PR curves, by comparing false positives with true positives rather than true negatives, avoid this issue in skewed data and provide a complementary metric to ROC curves. Figure 1 illustrates that the performance of LogitBoost clearly exceeds that of simple logistic regression in the mid-range of recall and precision.

Third, given the mixed results on which classifier works best in all cases, it seems worth highlighting when machine learning algorithms underperform or outperform logistic regression. In several scenarios, the performance of simple logistic regression was better than that of some machine learning algorithms. This finding is, to some extent, in accord with a few recent studies suggesting that traditional approaches to risk prediction (e.g., logistic regression) can be as effective as new machine learning approaches (Hamilton et al., 2014; Liu et al., 2011; Tollenaar & van der Heijden, 2013). However, it should be noted that the difference in performance between the best performing classifier and logistic regression was most pronounced when data were limited in terms of sample sizes, recidivism base rates, and number of predictors. For example, the AUC difference between random forests (0.814) and simple logistic regression (0.746) in the first-time sexual offending scenario was quite substantial.

Conclusion

There are a few limitations with this study that are worth noting. First, we evaluated the predictive performance of supervised learning algorithms on a sample of offenders released from Minnesota prisons. Despite the large sample size and considerable cross-sectional variation captured in the data, the findings may not be generalizable to other offender populations (e.g., probationers) or offenders from other jurisdictions. As such, assessing the algorithms we examined could yield different conclusions about predictive performance when applied to an offender sample from different populations, different jurisdictions, or different times. It is important to acknowledge that developing an actuarial tool that works effectively across time and place would be a daunting challenge if possible at all, and so would the validation of such tools be. Second, although we varied the parameters for each classification method in an effort

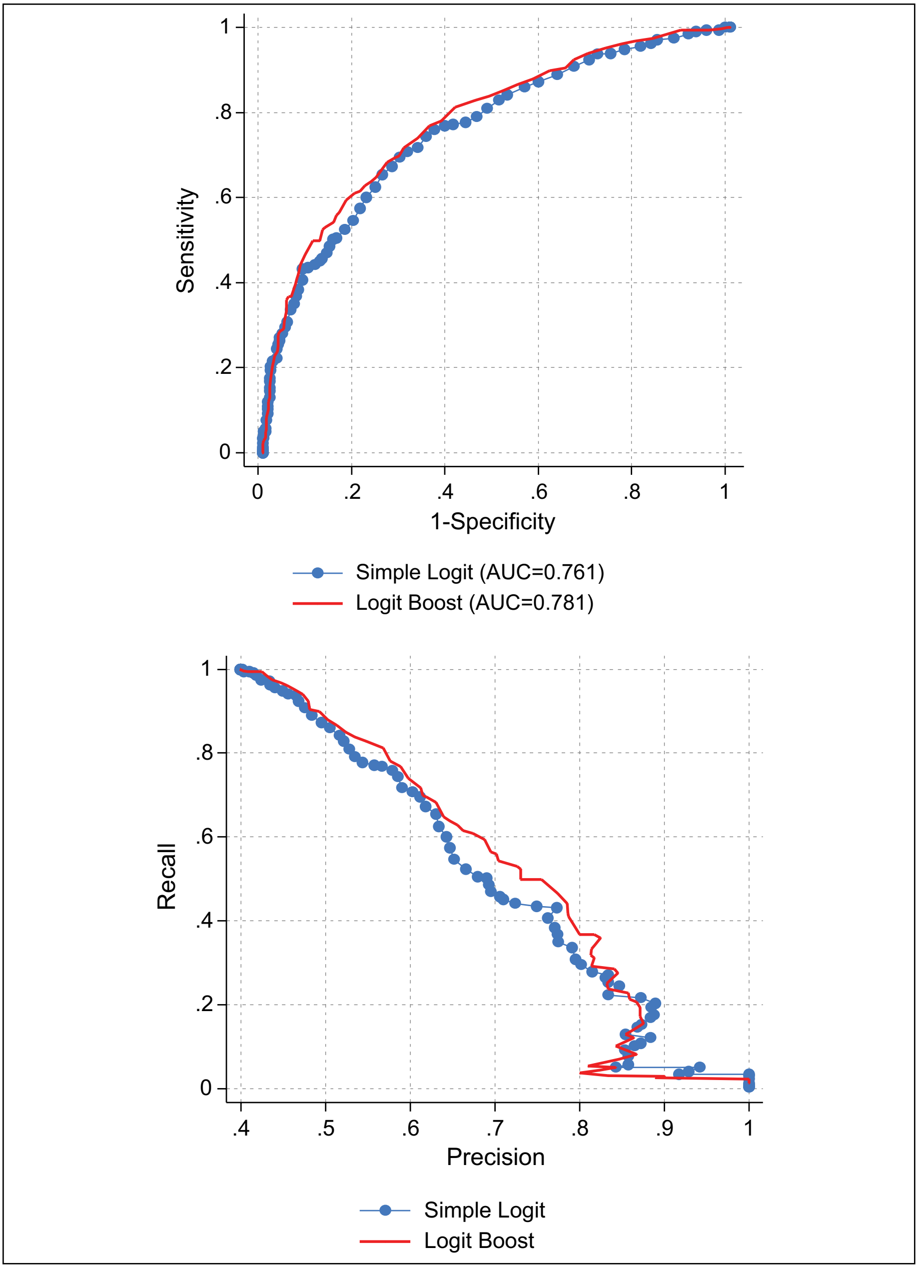

Figure 1. Comparison of AUC and PR curves.
Note. AUC = area under the curve; PR = Precision-Recall.

Table 3. Female General Recidivism: Random Forests and Logistic Regression by Risk Level.

	Random forests			Simple logistic regression		
Risk levels	Rate	Recidivists	Non-recidivists	Rate	Recidivists	Non-recidivists
Very high (top 20%)	82.3	130	28	73.4	116	42
High (next 20%)	48.7	77	81	48.7	77	81
Medium (middle 20%)	36.1	57	101	36.1	57	101
Low (bottom 40%)	16.2	51	264	20.6	65	250

to identify the best model, it is possible we may have missed the optimal tuning parameters for an algorithm within a given scenario.

Despite these limitations, we believe there are several implications that arise from the results presented in this study. First, the findings reported here and elsewhere suggest that, when properly tuned, the newer machine learning algorithms will, on average, achieve greater levels of predictive validity than older methods such as logistic regression, decision trees, or Naïve Bayes. Yet, given that our study also confirms the "no free lunch" theorem, reliance on a single algorithm—machine learning or otherwise—would likely limit the chances of developing the most accurate prediction instrument. Indeed, the value of testing multiple algorithms in search of the best one becomes more apparent when we consider the results from this study. For example, let us assume we had relied only on a traditional method such as simple logistic regression to develop a general recidivism prediction tool for female offenders. As shown above for female general recidivism, simple logistic regression accurately classified 69.7% of the offenders, its precision and recall rates were 0.716 and 0.400, respectively, and it had an AUC of 0.761. In comparison, random forests correctly classified 72.1% of the cases, it had values of 0.832 and 0.400 for precision and recall, and it had the highest AUC (0.789).

In practice, individual-level predicted probabilities from a risk assessment instrument are often aggregated into group-based risk levels to help prioritize offenders for programming and supervision-level decisions. For example, the MnSTARR classifies the top 20% as very high risk, the next 20% as high risk, the middle 20% as medium risk, and the bottom 40% as low risk. Offenders identified as very high and high risk (the top 40%) are prioritized for correctional programming in prison (Duwe, 2014a). If we aggregate the predicted probabilities from the logistic regression and random forests models according to those cut points, we see that random forests is more effective at identifying the highest risk offenders.

As shown in Table 3, the general recidivism rate for very high offenders is 82% for random forests and 73% for logistic regression. Although both algorithms have the same rate for high and medium risk offenders, the rate for low risk offenders is 16% for random forests versus 21% for logistic regression. Compared with the simple logistic regression model, the random forests model identified 14 more recidivists who would have been prioritized for prison-based programming that addressed their criminogenic needs.

Consistent with Ridgeway (2013b), we suggest that multiple approaches—machine learning algorithms in addition to more traditional methods such as logistic regression—should be considered when developing a recidivism risk assessment tool. To be sure, there will be instances in which logistic regression, for example, performs as well as one of the better machine learning algorithms such as random forests (e.g., male non-sexual violent recidivism). Unless multiple algorithms are assessed, however, it is not possible to make an a priori determination that a traditional method will achieve a level of predictive performance on par with a machine learning approach. Gaining an understanding of machine learning algorithms along with the various statistical and data mining software packages (e.g., R, WEKA, KNIME, Orange, Tanagra, Python, etc.) that support their use will, for most we imagine, entail a relatively steep learning curve (Barnes & Hyatt, 2012). Therefore, similar to the approach taken by Berk and Bleich (2013), researchers unfamiliar with machine learning may wish to concentrate their efforts by comparing the performance of older methods alongside the most promising machine learning algorithms—random forests, MultiBoosting, and LogitBoost.

Second, the decision about which method to use should depend on more than which one has the best predictive validity. It should also depend on how the risk assessment instrument will be used. For example, if the goal is to develop a second-generation risk assessment instrument designed to be administered only once to offenders that relies mostly on static, criminal history factors, the evidence shown here suggests one of the newer machine learning algorithms will likely produce the best instrument. Moreover, because one advantage common to many machine learning algorithms is that they can effectively filter out noisy data, we believe these methods would probably thrive in an environment in which the risk assessment process was automated and used a relatively large number of predictors drawn from a centralized database (i.e., similar to the instruments developed and used in Philadelphia). Nevertheless, it is worth emphasizing that even if only a small number of predictors are available, which is more likely to be the case in a manual risk assessment process, the findings suggest machine learning algorithms would still achieve relatively high levels of performance.

Third, if the goal is to develop a fourth-generation risk assessment instrument designed to be administered multiple times that attempts to incorporate dynamic factors, then the choice as to whether to use a machine learning algorithm becomes less clear. Most notably, there are practical challenges in using an algorithm that automatically considers nonlinear and/or interaction effects, as machine learning approaches typically do. To be sure, logistic regression can also account for nonlinear and interaction effects. In a main effects model, however, these types of effects can be removed. When a model contains either type of effect, nonlinear or interaction, it can produce individual-level results that may appear puzzling, particularly if the assessment is designed to be administered to the same offender more than once. For example, accumulating more institutional discipline convictions may actually decrease risk for some offenders rather than increase it. Conversely, completing programming in prison that, on the whole, mitigates recidivism risk may increase it for some offenders.

The challenge in using a model with nonlinear and/or interaction effects resides chiefly at the practitioner level. One of the long-standing concerns with machine learning algorithms is that they are black boxes or, at best, gray boxes, wherein it is difficult, if not impossible, to deduce how individual-level predictions are made. As Brennan and Oliver (2013) rightly pointed out, this relative lack of transparency in machine learning algorithms can make it problematic for practitioners, who frequently have to justify the decisions they make. Yet, the complexity of models that use interactions and nonlinear effects is also a major barrier to overcome when it comes to explanation and, ultimately, successful ongoing use of a risk assessment instrument.

We agree with Bushway (2013) that explaining individual-level predictions from a logistic regression model to practitioners can be difficult and becomes even more so when interaction and/or nonlinear effects are added. In fact, we suspect the original or modified Burgess methods are likely the only ones that can be easily interpreted by practitioners. We anticipate the explanation challenge can be overcome, to some extent, for most correctional risk assessment applications as long as the instrument is perceived as beneficial and consistent with existing theory and practice. For example, if an instrument "rewarded" prison misconduct (i.e., decreased an offender's risk) or "penalized" offenders (i.e., increased their risk) for completing risk-reduction programming (due to an interaction or nonlinear effect) to which they were directed, unintended consequences such as these would likely diminish support for its use.

Despite this dilemma, we suggest that fourth-generation risk assessment instruments based on machine learning algorithms could potentially improve correctional practice by better identifying the specific types of programming that would be most beneficial in reducing an individual offender's recidivism risk. For example, during the initial assessment, the instrument could identify the types of programming that would be most likely to lower an offender's risk. However, offenders could also be diverted from programming that would not reduce their risk (or may actually increase it). The challenge here, however, is not so much with the design and development of the instrument, but rather with its implementation.

We anticipate a fair amount of information technology (IT) expertise would be needed to successfully implement a fourth-generation risk assessment instrument based on a machine learning algorithm. After all, even for the second-generation tool they developed, Barnes and Hyatt (2012) acknowledged that its implementation required significant involvement from IT staff. In response to the reality that some jurisdictions lack the IT infrastructure to apply a fully automated, machine learning–based instrument, Berk and Bleich (2014) have recently recommended using classification trees as an alternative. However, if jurisdictions lack IT resources, it is unlikely they will have the wherewithal to develop their own risk assessment tool or have one created specifically for them (regardless of whether classification trees or some other method is used).

Another option for IT-deficient jurisdictions would be the development of a universal risk assessment instrument that is, similar to some of the widely used tools today (e.g., Level of Service/Case Management Inventory (LS/CMI), Correctional Offender Management Profiling for Alternative Sanctions (COMPAS), etc.), designed for use across multiple jurisdictions except that it would be based on a machine learning algorithm. Here again, IT expertise would be needed for the development of the application in which the instrument would operate, but its use would not depend on the IT capability of the jurisdiction. Because it would not be automated, such an instrument would need to be scored manually. Moreover, the instrument would require training and test data from multiple jurisdictions (e.g., city, county, state, country) to sufficiently establish validity, and inter-rater reliability data would need to be obtained to ensure that it could be scored consistently across raters.

The use of machine learning algorithms within criminal justice is still in its infancy. As such, there remains much to be understood about the limitations and challenges in using these methods for assessments of risk. Although an increase in the study and use of machine learning algorithms would require a greater commitment of time and resources from both researchers and criminal justice agencies, we expect the potential benefits would far outweigh the actual costs. Just as automating the risk assessment process significantly enhances the efficiency of correctional systems, more accurate predictions would further increase efficiency by improving the allocation of resources, which are almost always limited. Better allocation of resources, in turn, could also lead to improved recidivism outcomes, which would translate to decreased victimization and criminal justice system costs.

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Notes

1. Caruana and Niculescu-Mizil (2006) also evaluated regularized and unregularized models; however, in the unregularized models, we further assess performance by estimating main effects and interaction/nonlinear effects models.
2. Overfitting occurs when structural patterns discovered from one data set do not hold up in another. In other words, if a risk prediction model is excessively complex and captures nuanced patterns in the data, including noise, the prediction model may not perform as well when applied to new data where such particular patterns or relationships between predictors are not as strong or noise in the data looks different.
3. Boosting algorithms are susceptible to overfitting, which is why we used the available validation procedures (e.g., folds and resampling) in developing prediction models on the training set.
4. The notion of calibration can be understood in many ways, but it usually refers to two issues: (a) How error is distributed, and (b) how well calculated probabilities or confidence values approximate the rate at which the outcome actually occurs. The former is often discussed in the context of regression such that a model should be correct on average in the following sense:

$$E\left(Y|Y^{\wedge}\right)=Y^{\wedge}.$$

In other words, positive and negative errors should be distributed evenly and therefore cancel each other out. The latter concerns how well the predicted probabilities of the outcome approximate the actual probabilities, which is more germane to our discussion in this article. For example, if 3 in 10 offenders recidivate, 1 should expect the model to predict the outcome at the same rate (30%). As the predicted probabilities of the outcome are typically on a continuous scale, one way to calibrate the model is to change the threshold separating recidivists from non-recidivists. For practical reasons, 0.5 is conventionally used for this threshold and assumed as such in the example we discuss.

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